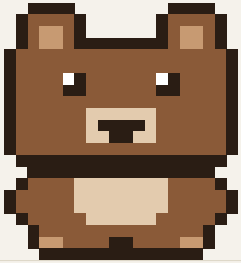
ROMA TRE · A.Y. 2026/27

Policy Evaluation Lab

Lecture 3: Econometrics tutorial



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Where we are

- Lecture 2: subsidised firms grew **6.39 points** faster than non-subsidised firms, but they were also larger, more export-oriented and more often in the South
- We know (because we simulated the data) that the true effect of the grant is **3 points**

Today, the first tool of the applied economist: **regression**

1. a regression on the treatment dummy reproduces the difference in means
2. adding **control variables** compares firms with the same observable characteristics
3. what we cannot control for creates **omitted-variable bias**

PART 1

Econometrics and regression



What is econometrics?

Econometrics is the application of mathematical statistics to economic data to lend empirical support to the models constructed by mathematical economics and to obtain numerical estimates (Samuelson, 1954).

In short, the job of an applied economist, or of a data analyst in a firm, is to:

- measure the relationship between variables (deduced or predicted by models)
- use such estimated relationships for predictions, forecasts and **decisions**: should the region keep funding the grant? should the firm extend the pilot to all its stores?

What is a regression?

- A **regression** allows you to estimate the impact of a variable of interest D on an outcome of interest Y, keeping other variables X constant
- Equation:

$$Y_i = \alpha + \gamma D_i + \beta_1 X_{1i} + \cdots + \beta_k X_{ki} + \epsilon_i$$

  - i denotes the unit of observation: here a **firm** (it could be a region, a store, a worker…)
  - Y is the **dependent variable** (outcome): e.g. sales growth
  - D is the **treatment**: e.g. the grant; X are **control variables**: e.g. size, export status
  - ϵ is the **error term**: everything else that affects Y

Adding the time dimension

The time dimension may be added to the regression model to account for changes over time:

$$Y_{it} = \alpha_t + \gamma D_{it} + \beta_1 X_{1it} + \cdots + \beta_k X_{kit} + \epsilon_{it}$$

- t might represent a year, quarter, month, day, etc.
- Observing the same unit i over time leads to **panel data**
- Different units sampled at different times result in **repeated cross-sections**

> **COMING UP**
>
> Panel data are the raw material of difference-in-differences: the same firms before and after a policy

PART 2

Data



Data and sample inference

Data collection:

- Ideally, you would observe every firm in the population of interest (e.g. all Italian manufacturing SMEs), but this is often not possible
- Researchers typically work with a **sample** of firms: from surveys, balance-sheet databases (e.g. AIDA, Orbis), administrative records of the agency that runs the policy
- Statistical methods are used to infer facts about the population from the sample

The sample should be:

- a **random draw** from your population of interest, to ensure representativeness

Data: cross-section

One observation per firm, at one point in time: the first rows of `firms_grant.csv`

firm_id	region	sector	age	exporter	employees_2015	sales_2015	grant
F0001	North	Food	15	0	16	1,826	0
F0002	South	Machinery	9	1	30	9,948	1
F0003	North	Furniture	22	1	10	2,763	0
F0004	Centre	Machinery	19	0	18	3,974	0
F0005	Centre	Textiles	18	1	34	7,343	0
F0006	Centre	Metal products	20	0	22	5,528	0
F0007	Centre	Furniture	13	0	30	3,116	0
F0008	South	Metal products	29	0	56	9,435	1

Simulated data. Sales in thousand euro.

Data: panel

The same firms observed in several years: one row per **firm-year**

firm_id	year	employees	sales	grant (D_{it})
F0001	2015	16	1,826	0
F0001	2019	17	1,911	0
F0002	2015	30	9,948	0
F0002	2019	30	11,280	1
F0003	2015	10	2,763	0
F0003	2019	10	3,194	0
F0004	2015	18	3,974	0
F0004	2019	18	3,723	0

The same data in "long" format. F0002 is treated only after the 2016 call: D_{it} changes over time.

PART 3

Ordinary least squares

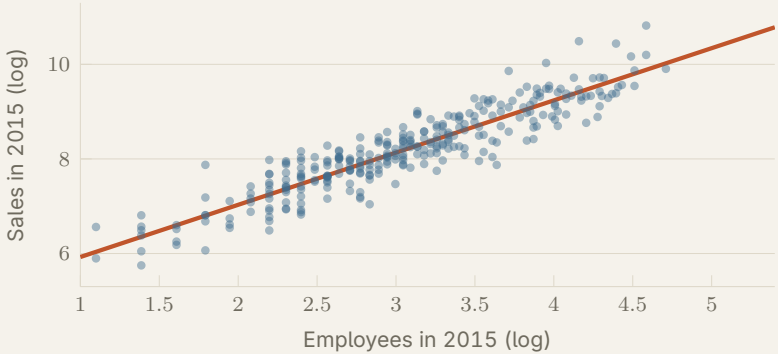


Scatter plot and regression

- A **scatter plot** visualises the relationship between two variables, showing how one variable changes as the other changes
- Regression analysis, particularly **Ordinary Least Squares (OLS)**, estimates the line that best describes this relationship
- The OLS line minimises the sum of the squared vertical distances between the observed values and the line (the squared **residuals**)

Example: are larger firms (more employees) also the ones that sell more? By how much?

Scatter plot and OLS fitted line: sales and employees



300 randomly drawn firms from `firms_grant.csv`. OLS line on all 1,500 firms: $\widehat{\log(\text{sales})} = 4.82 + 1.10 \log(\text{employees})$, $R^2 = 0.84$.

Reading the slope: logs and elasticities

$$\widehat{\log(\text{sales}_i)} = 4.82 + \mathbf{1.10} \times \log(\text{employees}_i)$$

- Both variables in logs: the slope is an **elasticity**
- A firm with 1% more employees sells, on average, about **1.1%** more
- Slightly more than 1%: in these data larger firms also sell more *per employee*
- $R^2 = 0.84$: size explains most of the differences in sales across firms

RULE OF THUMB FOR BUSINESS DATA

Sales, employees, wages, prices are skewed: take logs, and read coefficients as **percentage changes**

PART 4

Regression and the grant



A regression on a dummy is a comparison of means

Regress sales growth on the grant dummy only:

$$\text{SalesGrowth}_i = \alpha + \gamma\, \text{Grant}_i + \epsilon_i$$

- Average growth is α for non-subsidised firms and $\alpha + \gamma$ for subsidised firms
- So γ is the **difference in means** between the two groups

Estimates on our 1,500 firms (standard errors in parentheses):

$$\widehat{\text{SalesGrowth}}_i = \underset{(0.26)}{3.37} + \underset{(0.51)}{6.39}\, \text{Grant}_i$$

> Exactly Table 1 of Lecture 2, selection bias included

Adding control variables

We add the characteristics that were unbalanced in Table 2 of Lecture 2:

$$\begin{aligned}\text{SalesGrowth}_i = \alpha + \gamma\,\text{Grant}_i + \beta_1 \log(\text{Empl}_i) + \beta_2 \text{Exporter}_i \\ + \beta_3 \text{Age}_i + \beta_4 \text{South}_i + \beta_5 \text{Family}_i + \epsilon_i\end{aligned}$$

- γ is now the difference in growth between subsidised and non-subsidised firms **with the same** size, export status, age, location and ownership
- "Controlling for X" $=$ comparing firms that look alike in X
- This deals with the imbalances we found in Table 2 of Lecture 2, **if** those are the only differences between the two groups

Results

Dep. var.: sales growth 2015–19 (%)	(1)	(2)
Grant	**6.389***** (0.514)	**5.453***** (0.522)
log(Employees)		1.587*** (0.313)
Exporter		2.736*** (0.504)
Age (years)		−0.076*** (0.018)
South		−1.828*** (0.485)
Family-owned		−0.889* (0.532)
Constant	3.370*** (0.264)	1.119 (1.153)
N	1,500	1,500
R^2	0.10	0.16

Robust standard errors in parentheses. $^{***}\,p<0.01$, $^{**}\,p<0.05$, $^{*}\,p<0.1$.

- With controls, the grant coefficient falls from 6.39 to **5.45**: part of the naive gap was due to size, exports and location
- But 5.45 is still far from the true effect of **3**

Interpreting the coefficients of column (2)

Each coefficient: the change in Y for a one-unit change in that variable, **holding the others constant**

- **Grant** $= 5.45$: subsidised firms grew 5.45 points faster than comparable firms without the grant
- **Exporter** $= 2.74$: exporters grew 2.7 points faster than similar non-exporters
- **Age** $= -0.08$: ten more years of age, 0.8 points less growth
- **South** $= -1.83$: southern firms grew 1.8 points less than similar firms elsewhere
- **log(Employees)** $= 1.59$: a firm 10% larger grew about $1.59 \times 0.10 \approx 0.16$ points faster

Careful: we evaluate the grant; the other coefficients are correlations, not necessarily causal effects

PART 5

Omitted-variable bias



Why is 5.45 not 3?

- In Lecture 2 we said that what makes Alfa apply is rarely in the balance sheet: **management quality**
- Management quality (MQ) affects growth **and** the probability of getting the grant
- It is not in `firms_grant.csv`, so we cannot control for it: it stays in the error term ϵ_i

> A variable that affects Y, is correlated with D, and is left out of the regression creates **omitted-variable bias** (OVB)

Since we simulated the data, we can do what no real analyst can: add MQ to the regression and see what happens.

The “long” regression

Dep. var.	**Sales growth 2015–19 (%)**	
	(2) Short	(3) Long
Grant	5.453***	2.715***
	(0.522)	(0.514)
Management quality		3.819***
		(0.238)
Controls of column (2)	Yes	Yes
N	1,500	1,500
R^2	0.16	0.28

Robust standard errors in parentheses. Management quality is standardised (mean 0, s.d. 1) and is **not** in the students’ dataset.

- With MQ, the grant coefficient drops to **2.7**, close to the true 3
- One standard deviation of MQ is worth 3.8 points of growth: more than the grant itself

The omitted-variables bias formula

$$\underbrace{\gamma^{short}}_{5.45} = \underbrace{\gamma^{long}}_{2.71} + \underbrace{\beta_{MQ}}_{3.82} \times \underbrace{\delta}_{0.72}$$

- β_{MQ}: effect of the omitted variable on the outcome (in the long regression)
- δ: difference in the omitted variable between treated and controls, holding the other X fixed (regression of MQ on Grant and the controls)
- $3.82 \times 0.72 = 2.74$: the omitted-variable bias. $2.71 + 2.74 = 5.45$

> Short = Long + (Effect of omitted on Y) $\times$ (Regression of omitted on D)

The sign of the bias

	Omitted var. **more** common among treated ($\delta > 0$)	Omitted var. **less** common among treated ($\delta < 0$)
Raises Y ($\beta > 0$)	**upward** bias	downward bias
Lowers Y ($\beta < 0$)	downward bias	**upward** bias

- Our grant: MQ raises growth and is higher among subsidised firms $\Rightarrow$ **upward** bias
- Rescue aid: the omitted "crisis" lowers growth and is more common among the treated $\Rightarrow$ **downward** bias

IN REAL DATA

We never observe MQ: we can only **argue** about the sign of the bias, or use a design that removes it (experiments, natural experiments)

PART 6

Standard errors and significance



Standard errors and significance (I)

- The **standard error** (in parentheses) measures how precisely a coefficient is estimated: another sample of firms would give a somewhat different number
- We want to know whether we can reject the **null hypothesis** $H_0 \colon \gamma = 0$ (the grant has no effect)
- In large samples, if $|\hat{\gamma}/s.e.| > 1.96$ we reject H_0 at the **5% level**; > 2.58 at the **1% level**

Column (2): $t = \dfrac{5.453}{0.522} = 10.45 > 2.58 \Rightarrow$ significant at 1% (***)

PRECISE IS NOT THE SAME AS UNBIASED

The standard error measures chance, not bias: 5.45 is very precisely **wrong**

Standard errors and significance (II)

- 1.96 and 2.58 are critical values of the standard normal distribution, to which Student's t distribution converges in large samples
- *** tells you that we can reject the null hypothesis that the coefficient is equal to zero at the 1% significance level (if H_0 were true, the probability of wrongly rejecting it would be at most 1%)
- ** and * usually refer to the 5% and 10% significance levels (95% and 90% confidence)

> *** means **statistically significant at the 1% level**, not a "99% chance" that $\gamma \neq 0$

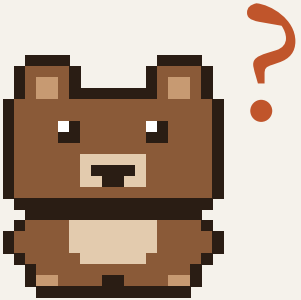
Robust and clustered standard errors

- **Robust** standard errors allow the spread of the errors to differ across firms (*heteroskedasticity*): large firms' growth is less volatile than small firms'. They are the default in applied work, and the ones in our tables
- **Clustered** standard errors allow the errors of firms in the same group to be correlated: firms in the same province or sector share demand shocks, local banks, the same regional call
- Rule: cluster at the level at which the treatment is assigned (e.g. by region, if the grant is decided region by region)

> Clustering usually makes standard errors **larger**: ignoring it overstates precision

Next steps

- **Friday, Lab 1**: first steps in R, on `firms_grant.csv`. Bring a laptop with R and RStudio
- **Lab 2**: today's table in R, with `lm()`
- **Next**: randomised experiments. If the grant were assigned by lottery, MQ would be the same, on average, in both groups: $\delta = 0$, and the omitted-variable bias disappears



Questions?

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